

SARA CHOPRA

Hamburg , Germany || Science Communication & Physics

Email Address: sarachopra2001@gmail.com

LinkedIN : <https://www.linkedin.com/in/sara-chopra/>

Portfolio: sarachopra2001.github.io

GitHub: github.com/SaraChopra2001

Physicist and science communicator with a background in theoretical particle physics and research-driven public outreach. Experienced in translating complex scientific concepts into accessible visual and digital formats, including interactive web explainers and multimedia content. Currently developing web-based communication projects for the Cluster of Excellence “Advanced Imaging of Matter” (CUI) while expanding skills in machine learning and AI-assisted workflow development. Seeking roles that combine scientific rigour with communication, digital & design strategy, and AI-assisted workflows.

Professional Experience

Student Assistant (PR & Communication Department)

03/2024 - 03/2026

Cluster of Excellence for Advanced Imaging (CUI), Universität Hamburg

- Managing social media content and strategy for varied audiences across Instagram, LinkedIn, and web, including research interview and lab tour video production (ideation to production to editing) and supported outreach events on-site, eg. Science City Day.
- Designing outreach assets for events and digital platforms (posters, graphics, reels/videos, print materials).(See samples here)
- Provide structured office + analytical support: maintain internal datasets for projects along with performed event feedback evaluation.
- Gave a lecture to high school students on “Particle and Accelerator Physics” for Lights and Schools.
- Recently, contributed to a CUI news piece for a latest research publication. Currently developing a CUI web infographic explaining the different research areas.

Independent Project: CMS Detector Web Infographic
Scrollytelling webpage explaining detector components and physics. Built with HTML/CSS/JS; used AI assistance to speed up implementation; content inspired by CERN/CMS public resources. (10 day build)

Communication & Media

- Science communication for specialist & non-specialist audiences.
- Social media content + strategy
- Short-form video production/editing + interview coordination

Design & Digital

- Canva, Adobe Creative Suite
- Blender (for 3D scientific visualisation)
- MS Office
- Web basics: HTML/CSS

Languages

English: Fluent (Native level),
German: Level (B2.1) Intermediate,
Hindi: Native

Conferences & Workshops

- KMBP School on QFT, Higher Topological Algebra, and Gauge Symmetry (2024)
- DPG (Deutsche Physikalische Gesellschaft) Conference, 2024
- NIUS Camp (Homi Bhabha Centre for Fundamental Research)
- Science Communication Workshop, Humboldt University (Dr. Alison Pinto, 2024)

Scholarships/Awards

- Merit Scholarship for International Students: 2024 - 2025 (Universität Hamburg)
- 50% Academic Fee Scholarship in Bachelor's: 2019 & 2021 (Amity Univeristy)
- Goethe Institute: Scholarship Program (Sri Lanka): June 2017

Education

10/2022 - 04/2026

Universität Hamburg,
Germany

Masters in Physics

Accelerator and Particle
Physics (specialisation)

Master Thesis Project :

‘Sum Rules in Extended Higgs
Model and their Phenomenological
Applications

Master’s Result: (German System)
(Grade Range 1-6 (1- highest grade))

Final Result : 1.29

2019 - 2022

AIAS, Amity University,
Noida, India

B.Sc. (Honors) Physics

Bachelor’s Result: (Indian Grading System)
(10 is the highest grade)

1st Year: 9.52/10
2nd Year: 9.38/10
3rd Year: 9.30/10

2005 - 2019

Springdale’s School, Pusa
Road, New Delhi, India

Higher Secondary Schooling

High School Result: (Indian Grading System)
(100% is the highest grade)

12th Standard: 91.5%
Science Stream with Economics & German
10th Standard: CGPA: 10/10

Technical & Analytical Skills

- Specialised in Theoretical and Phenomenological particle Physics & Computational Analysis.
- Programming & Data Analysis
- Scientific Writing & Documentation

Technical Software Used:

Mathematica (FeynCalc, FeynArts, SARAH), Python, SQL, LaTeX.

Volunteering & Extracurricular Activities

- Volunteer at eVidyaloka (2022): Taught and created educational content for students in rural areas.
- Each One Teach Many Program (2014-2019): Provided education to eight underprivileged children.

Hobbies and Interests

- **Art & Design:** Sketching, acrylic & oil painting
- **Media & Storytelling:** Video editing, science interviews
- **Music, Cooking, Travel:** Engaging with cultures and creative expression
- **Psychology and Neuroscience**

Research Experience

Master Thesis

07/2023 - 04/2026

Sum Rules in Extended Higgs Models and Their Phenomenological Applications, Institute of Theoretical Physics II, Universität Hamburg

Supervisors: Dr. Gudrid Moortgart Pick and Dr. Sven Heinemeyer

- Conducted theoretical and phenomenological analysis of extended Higgs sector models (2HDM, Georgi Machacek, Septet).
- Applied Mathematica and Python packages to evaluate phenomenological constraints and to conduct comparative analysis.
- Compared theoretical predictions and consistency of BSM models to collider observables at the LHC and computing allowed and excluded cross sections using HiggsTools.

Research Internship: Humboldt University, Berlin (2024)

10/2024 - 01/2025

Under Dr. Olaf Hohm (RTG 2575: Rethinking Quantum Field Theory)

- Studied homotopy transfer and BV algebras for gauge-invariant formulations of the path integral; extended the BV formalism to 4-term complexes with gauge symmetries.
- Investigated Chevalley–Eilenberg algebras and their role in deriving the Faddeev–Popov determinant; participated in weekly seminars on higher algebra and QFT.

Volunteering and Leadership Experience

Vice President and Member:

2019 - 2022 (Member 3 years, VP: 2020 - 2021)

KugelBlitz Physics Club, Amity University

- Led the transition from offline to online events, successfully implementing activities like research discussions, virtual treasure hunt, and board games.
- Established the club's organizational structure and social media presence, implementing a strategy that increased event audience from 5 to 50-75 participants.
- Acquired sponsorships and expanded active membership from 10 to 25, strengthening community engagement.
- Organised 20+ events and led brainstorming sessions, fostering a culture of innovation and participation.

Content & Design Team Member

04/2020 - 09/2020 (6 months)

Volunteer at NGO: GirlUp Empower

- Created over 10 articles on topics like women in science, contemporary beauty standards, and refining research and writing skills.
- Designed around 5-10 social media posts weekly, ensuring visually appealing and informative content for a broader audience.
- Developed an understanding of aesthetics, creative storytelling, and audience engagement through marketing-focused design.



TRANSCRIPT OF RECORDS

Sara Chopra

University of Hamburg

20148 Hamburg

Germany

Tel.: +49 40 42838-0

Fax: +49 40 42838-6594

**Faculty of Mathematics, Informatics and
Natural Sciences**

Desired degree: Master of Science
Physics

The student has completed the degree.

Last Name: Chopra

First Name: Sara

Date of Birth: 1 April 2001

Place of Birth: Delhi

Enrolled on: 1 October 2022

Student ID No.: 7597332

Number/Type	Name	Semester/Date	ECTS Credits	Grade
Physics (M.Sc.)			121	1,29
Advanced Master's Courses			49	
PHY-MV-BE-E26	Advanced particle physics*		8	1,3
ÜB	Exercises in advanced particle physics*	Winter 22/23		
VL	Advanced particle physics*	Winter 22/23		
Final module exam(s): Block exam				1,3
PHY-MV-BE-E17	Seminar: Physics and application of laser-plasma accelerators*		3	1,0
SEM	Seminar: Physics and application of laser-plasma accelerators*	Winter 23/24		
Final module exam(s): Final exam				1,0
PHY-MV-BE-T12	Phenomenology of physics beyond the standard model*		6	1,0
ÜB	Exercises in phenomenology of physics beyond the standard model*	Winter 23/24		
VL	Phenomenology of physics beyond the standard model*	Winter 23/24		
Exam	Final exam			1,0
PHY-MV-BE-T02	Physics of the standard model*		6	1,0
ÜB	Exercises in physics of the standard model*	Summer 23		
VL	Physics of the standard model*	Summer 23		
Final module exam(s): Final exam			6 Jul 2023	1,0
PHY-MV-BE-T04	Quantum field theory I		8	1,3
ÜB	Exercises in quantum field theory I*	Winter 22/23		
VL	Quantum field theory I*	Winter 22/23		
Final module exam(s): Block exam			29 Mar 2023	1,3
PHY-MV-FN-T25	Symmetry groups in physics*		8	1,0
ÜB	Exercises in symmetry groups in physics	Summer 23		
VL	Symmetry groups in physics	Summer 23		
Final module exam(s): Final exam			8 Aug 2023	1,0

Number/Type	Name	Semester/Date	ECTS Credits	Grade
PHY-MV-FN-E39	Methods in nanobiotechnology I*		7	1,7
PR	Practical: methods in nanobiotechnology I*	Winter 23/24		
ÜB	Exercises in methods in nanobiotechnology I*	Winter 23/24		
VL	Methods in nanobiotechnology I*	Winter 23/24		
Final module exam(s): Module exam				1,7
PHY-MV-BP-E05	Seminar on biomedical physics I*		3	2,7
SEM	Seminar on biomedical physics*	Winter 22/23		
Exam	Presentation and paper			2,7
Complementary Subject			12	
SPLV	German - "B1.2"	Summer 23	6	b
Exam	Block exam			1,7
SPLV	German Intensive Course - "B2.1"	Winter 23/24	6	b
Exam	Block exam			2,0
Research Phase			30	
PHY-MF-BE-EP	Introductory project (Accelerator physics and particle physics)		15	b
Final module exam(s): Final assignment				b
PHY-MF-BE-VPb	Preparatory project (Accelerator physics and particle physics)		15	b
Final module exam(s): Final assignment				b
Master thesis			30	
PHY-MF-MA	Master's thesis		30	1,3
Final Paper		8 Oct 2025		1,3
Master's thesis:	Sum Rules for the Extended Higgs Sector Models: A comparative analysis of 2HDM, GM and the Septet Model*			
Examiners:	Prof. Dr. Gudrid Moortgat-Pick, Prof. Dr. Sven Heinemeyer			
Oral exam		9 Oct 2025		1,3

Number/Type	Name	Semester/Date	ECTS Credits	Grade
Additional			0	
SPLV	German - "B1.1"	Winter 22/23	6	1,7
Exam	Block exam			1,7
Current grade			excellent	1,29

A minimum of 120 ECTS credits is required to successfully complete the program.

Hamburg, 20 January 2026

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Note: This document includes only successfully completed courses and examinations.

Grading system per component:

1,0 / 1,3	=	excellent
1,7 / 2,0 / 2,3	=	good
2,7 / 3,0 / 3,3	=	satisfactory
3,7 / 4,0	=	sufficient
5,0	=	insufficient

Calculations for the final grade and for individual components are determined by departmental regulations.

* = in the original language

b = pass

inc. = module/course incomplete

e = successfully completed



TRANSCRIPT OF RECORDS

Sara Chopra

Universität Hamburg

20148 Hamburg
Deutschland

Tel.: +49 40 42838-0
Fax: +49 40 42838-6594

**Fakultät für Mathematik, Informatik und
Naturwissenschaften**

Angestrebter Abschluss: Master of Science
Physics

Das Studium gilt als bestanden.

Nachname: Chopra
Vorname: Sara

Geburtsdatum: 1. April 2001
Geburtsort: Delhi

Immatrikuliert seit: 1. Oktober 2022
Matrikelnummer: 7597332

Nummer/Typ	Name	Semester/Datum	ECTS Credits	Note
Physics (M.Sc.)			121	1,29
Advanced Master's Courses			49	
PHY-MV-BE-E26	Advanced particle physics		8	1,3
ÜB	Exercises in advanced particle physics	WiSe 22/23		
VL	Advanced particle physics	WiSe 22/23		
Modulabschlussprüfung(en): Blockprüfung				1,3
PHY-MV-BE-E17	Seminar: Physics and application of laser-plasma accelerators		3	1,0
SEM	Seminar: Physics and application of laser-plasma accelerators	WiSe 23/24		
Modulabschlussprüfung(en): Modulabschlussprüfung				1,0
PHY-MV-BE-T12	Phenomenology of physics beyond the standard model		6	1,0
ÜB	Exercises in phenomenology of physics beyond the standard model	WiSe 23/24		
VL	Phenomenology of physics beyond the standard model	WiSe 23/24		
Prüfung	Modulabschlussprüfung			1,0
PHY-MV-BE-T02	Physics of the standard model		6	1,0
ÜB	Exercises in physics of the standard model	SoSe 23		
VL	Physics of the standard model	SoSe 23		
Modulabschlussprüfung(en): Modulabschlussprüfung			06.07.2023	1,0
PHY-MV-BE-T04	Quantenfeldtheorie I		8	1,3
ÜB	Exercises in quantum field theory I	WiSe 22/23		
VL	Quantum field theory I	WiSe 22/23		
Modulabschlussprüfung(en): Modulabschlussprüfung			29.03.2023	1,3
PHY-MV-FN-T25	Symmetry groups in physics		8	1,0
ÜB	Übung zu Symmetriegruppen in der Physik	SoSe 23		
VL	Symmetriegruppen in der Physik	SoSe 23		
Modulabschlussprüfung(en): Modulabschlussprüfung			08.08.2023	1,0

Nummer/Typ	Name	Semester/Datum	ECTS Credits	Note
PHY-MV-FN-E39	Methods in nanobiotechnology I		7	1,7
PR	Practical: methods in nanobiotechnology I	WiSe 23/24		
ÜB	Exercises in methods in nanobiotechnology I	WiSe 23/24		
VL	Methods in nanobiotechnology I	WiSe 23/24		
Modulabschlussprüfung(en): Modulabschlussprüfung				1,7
PHY-MV-BP-E05	Seminar on biomedical physics I		3	2,7
SEM	Seminar on biomedical physics	WiSe 22/23		
Prüfung	Referat und Hausarbeit			2,7
Complementary Subject			12	
SPLV	DaF - "B1.2"	SoSe 23	6	b
Prüfung	Blockprüfung			1,7
SPLV	DaF-Intensivkurs - B2.1	WiSe 23/24	6	b
Prüfung	Blockprüfung			2,0
Research Phase			30	
PHY-MF-BE-EP	Einarbeitungsprojekt (Beschleuniger- und Elementarteilchenphysik)		15	b
Modulabschlussprüfung(en): Projektabschluss				b
PHY-MF-BE-VPb	Vorbereitungsprojekt (Beschleuniger- und Elementarteilchenphysik)		15	b
Modulabschlussprüfung(en): Projektabschluss				b
Master thesis			30	
PHY-MF-MA	Masterarbeit Physik		30	1,3
Abschlussarbeit		08.10.2025		1,3
Thema:	Sum Rules for the Extended Higgs Sector Models: A comparative analysis of 2HDM, GM and the Septet Model			
Gutachtende:	Prof. Dr. Gudrid Moortgat-Pick, Prof. Dr. Sven Heinemeyer			
Kolloquium		09.10.2025		1,3

Nummer/Typ	Name	Semester/Datum	ECTS Credits	Note
Zusätzliche Leistungen			0	
SPLV	DaF - "B1.1"	WiSe 22/23	6	1,7
Prüfung	Blockprüfung			1,7
Gleitende Gesamtnote		sehr gut		1,29

Es sind 120 ECTS-Credits zu erbringen.

Hamburg, 20. Januar 2026

Dieses Dokument ist ohne Unterschrift gültig.

Hinweis: Dieses Dokument enthält nur bestandene Prüfungsversuche.

Bewertungssystem Teilleistungen:

1,0 / 1,3	=	sehr gut
1,7 / 2,0 / 2,3	=	gut
2,7 / 3,0 / 3,3	=	befriedigend
3,7 / 4,0	=	ausreichend
5,0	=	ungenügend

Die Art der Berechnung zur Ermittlung der Gesamtnote und der Teilleistungsnote wird in den Fachspezifischen Bestimmungen des Faches geregelt.

b = bestanden

uv. = Die Studienleistung ist noch nicht vollständig abgeschlossen.

e = erfolgreich erbracht

PORTFOLIO

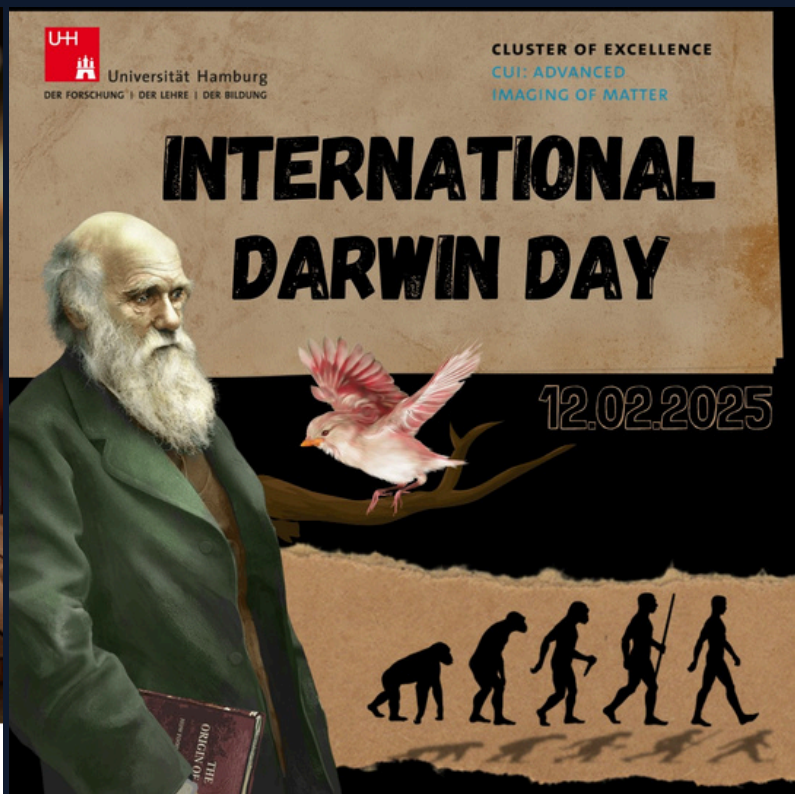
2026

Sara Chopra

CUI: Instagram Posts

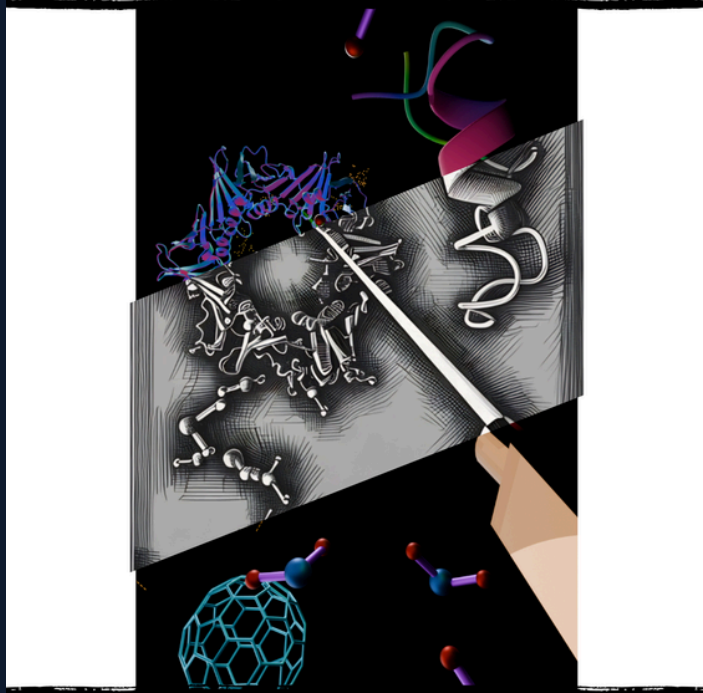


CUI: Instagram Posts



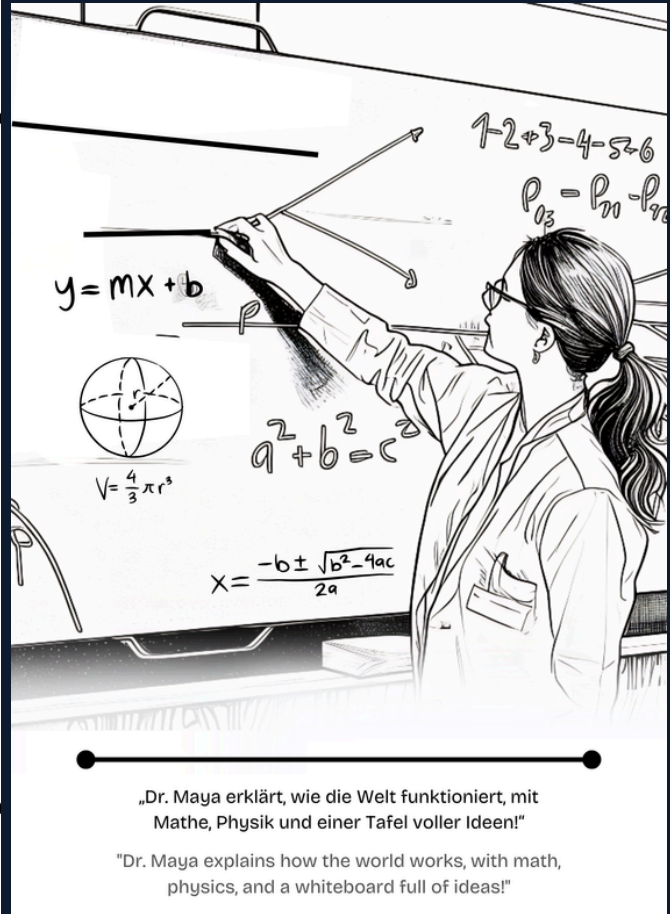
Other design samples

CUI: MALEBUCH



CLUSTER OF EXCELLENCE
CUI: ADVANCED
IMAGING OF MATTER

U+H
Universität Hamburg
DER FORSCHUNG | DER LEHRE | DER BILDUNG



CUI: Research Slambook



CUI: SEITE DER GRUPPE

GROUP INFORMATION

Gruppenfoto, Illustration/Zeichnung der Forschung

PRINCIPLE INVESTIGATOR _____

GRUPPENMITGLIEDER _____

HAUPTFORSCHUNGSBEREICHE _____

WAS IST IHR DENKWÜRDIGSTER CUI-MOMENT?

INTERESSANTE ERGEBNISSE/ WICHTIGE MEILENSTEINE

AKTUELLE FORSCHUNG ZIELE UND METHODIK

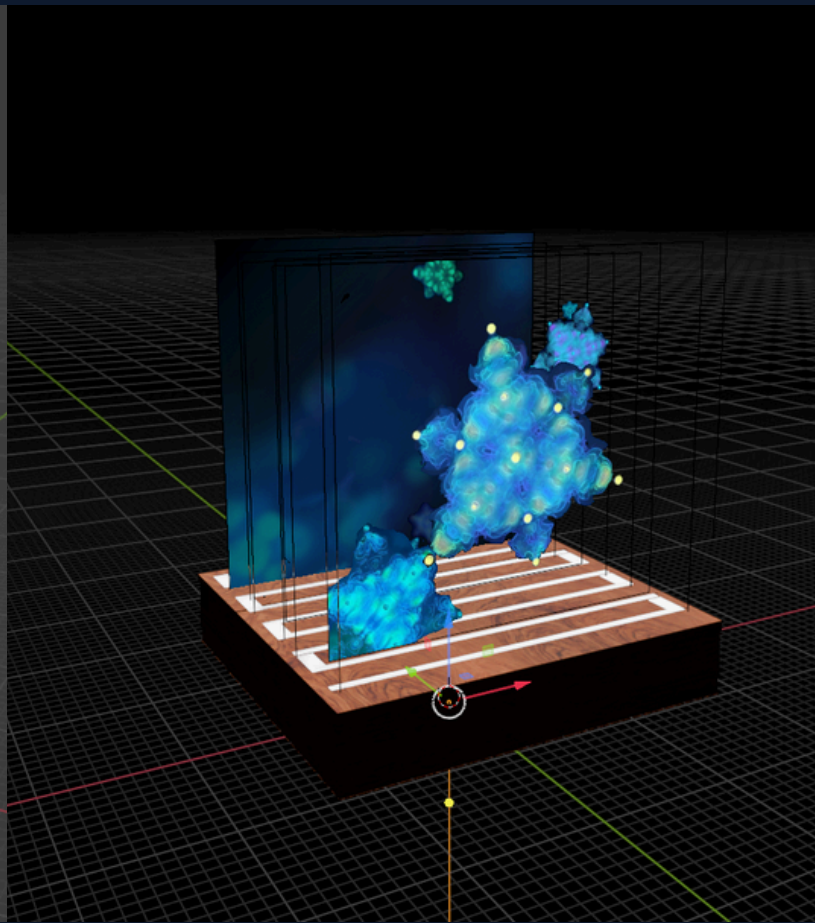
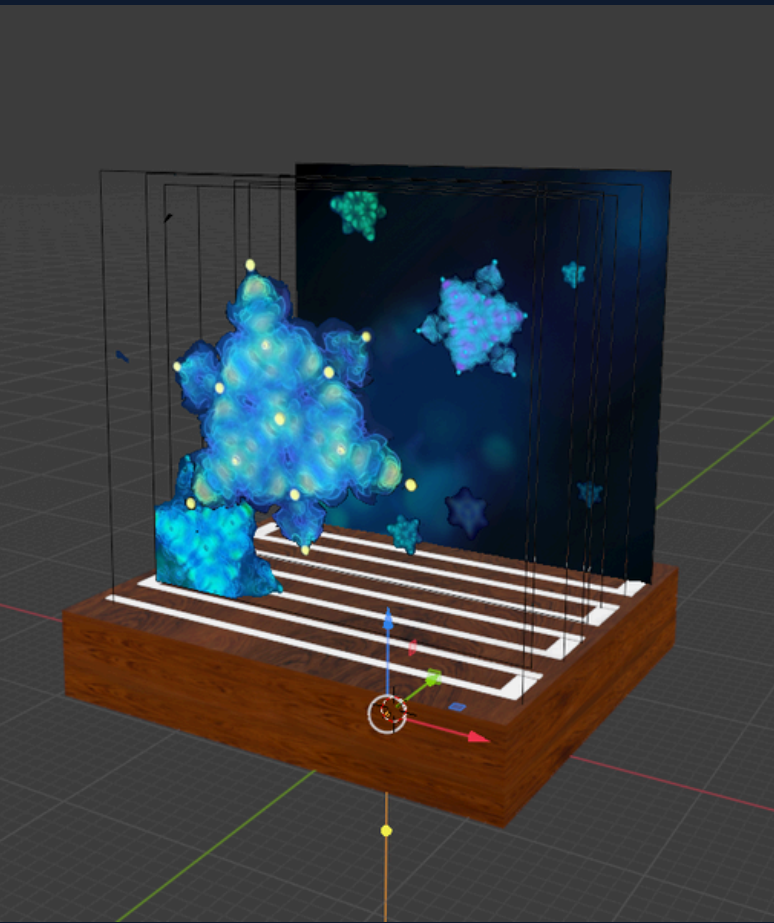
WAS IST DAS BESONDERE AN IHRER ARBEIT?

WAS IST IHR CLUSTER-HIGHLIGHT DER LETZTEN 7 JAHRE?

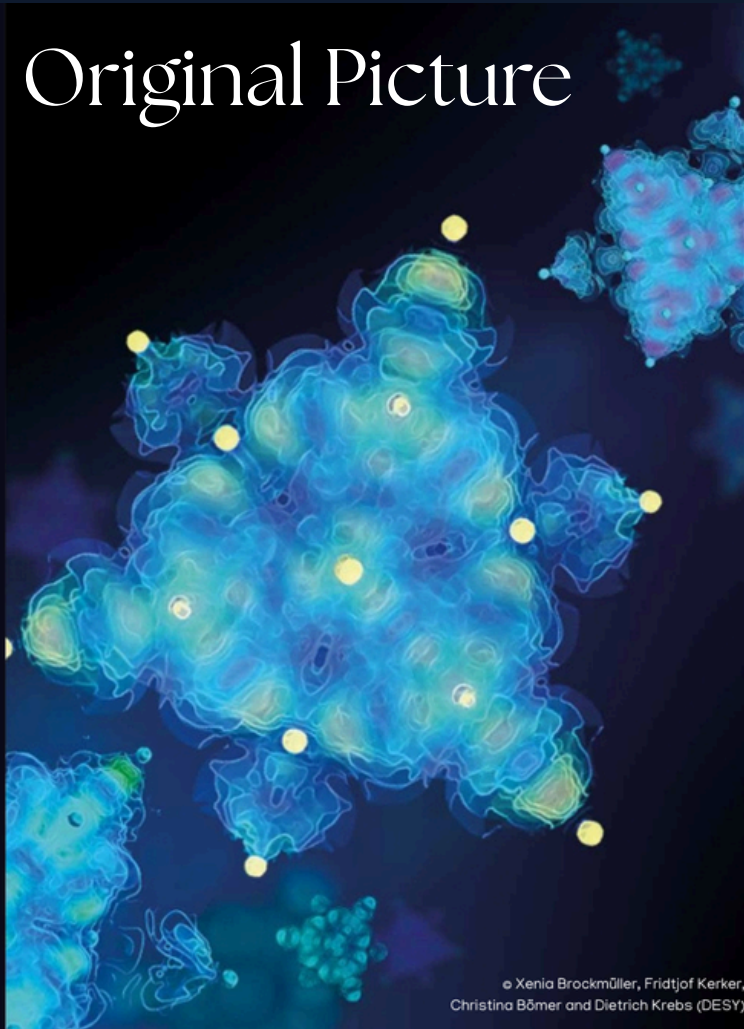
U+H
Universität Hamburg
DER FORSCHUNG | DER LEHRE | DER BILDUNG

CLUSTER OF EXCELLENCE
CUI: ADVANCED
IMAGING OF MATTER

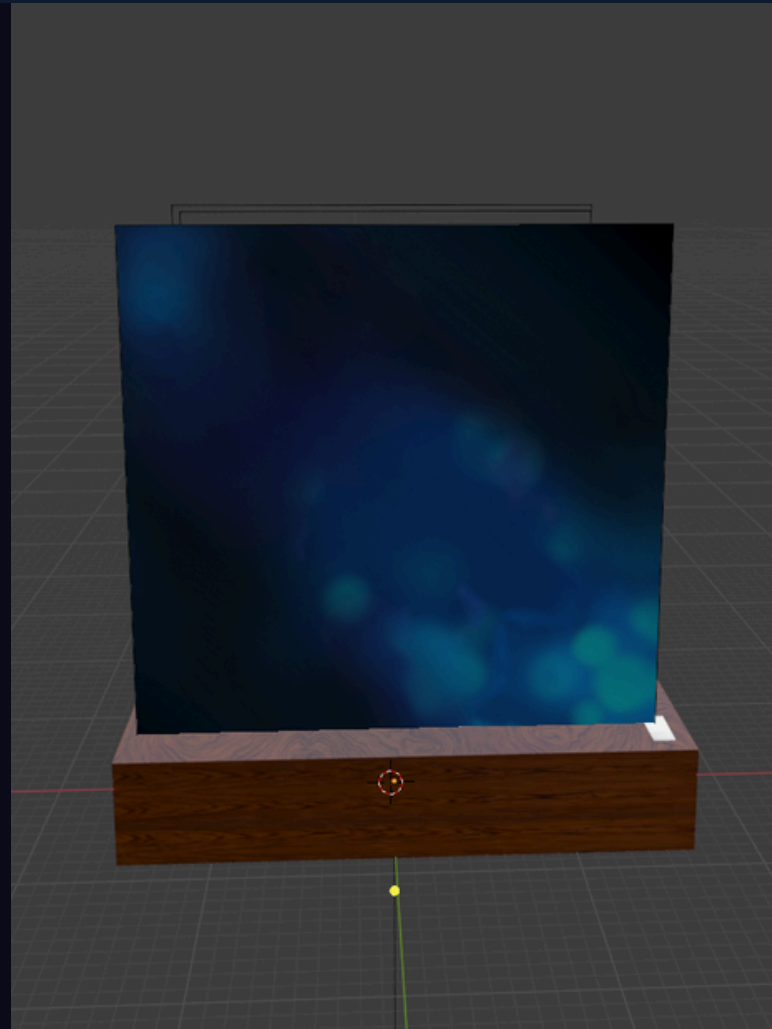
3D Model



Original Picture



© Xenia Brockmüller, Fridtjof Kerker,
Christina Bömer and Dietrich Krebs (DESY)



Physics Club (AIAS) Events



BEYOND GRAVITY

PHYSICS & Board Game

KUGELBLITZ Presents

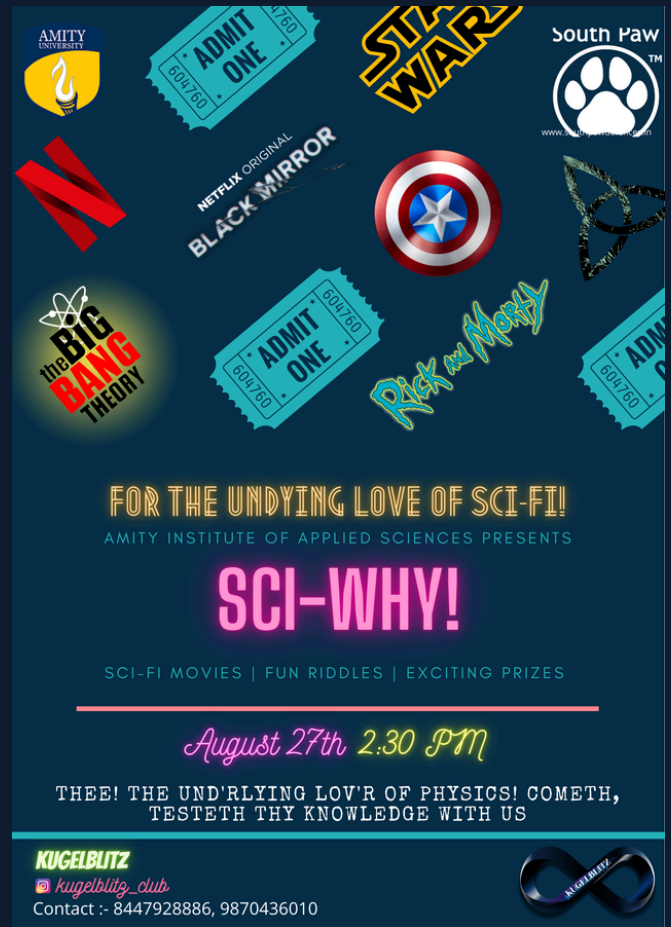
READY SET GO!!!

FLIPPITY

OCT 22
3:00 PM

@KUGELBLITZ_CLUB
CONTACT: 9870436010
8447928886

South Paw



AMITY UNIVERSITY

South Paw

ADMIT ONE

NETFLIX ORIGINAL **BLACK MIRROR**

STAR WARS

CAPTAIN AMERICA

the BIG BANG THEORY

Rick and Morty

ADMIT ONE

ADMIT ONE

FOR THE UNDYING LOVE OF SCI-FI!

AMITY INSTITUTE OF APPLIED SCIENCES PRESENTS

SCI-WHY!

SCI-FI MOVIES | FUN RIDDLES | EXCITING PRIZES

August 27th 2.30 PM

THEE! THE UND'RLYING LOV'R OF PHYSICS! COMETH,
TESTETH THY KNOWLEDGE WITH US

KUGELBLITZ
@kugelblitz_club
Contact :- 8447928886, 9870436010



AMITY UNIVERSITY

South Paw

AIAS & SOUTH PAW

KUGELBLITZ PRESENTS

PHYSICS QUEST

A HUNT FOR THE TREASURES OF PHYSICS

A QUEST THROUGH THE REALM OF PHYSICS

SEPTEMBER 24TH
3:00 PM

@KUGELBLITZ_CLUB

CONTACT: 9870436010, 8447928886

GirlUp Empower

The Truth About Geishas:

Artists or Prostitutes?



Mizuage (Virginity of Geisha)

A Geisha's Mizuage was sold by the Geisha House to Sponsor her expensive Training and Kimonos.



BURGER MENU

This is a
Burger!!



This is
not Meat!!



"HEME" is what makes these plant based meat,
FEEL like MEAT



SIDES
CHEESE FRIES



IT IS EXTRACTED FROM SOYA PLANTS
& CAUSES THE EXPLOSION OF FLAVOUR & SIZZLE

Torches Of Freedom



MARLBORO
America's Finest Cigarette



girlupempower · Follow



girlupempower Edited · 252w

Smoking is a big problem these days, right? There are women, men and even children from all over the world who're addicted to smoking. Since smoking by women has been historically disapproved of, how did they get into it? To understand this, we have to go back to the first World War, when gender roles were diluted due to necessity. With men away at war, women were forced to move out of their homes and enter the workplace. Feminism became huge at this time. Women were demanding equal



56 likes

July 10, 2020

Log in to like or comment.

Digital Artwork and Paintings

